

Computational Optimal Transport: LP, Regularization, and the Gaussian Case

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1 Problem setup

Let $X = (x_i)_{i=1}^m \subset \mathbb{R}^2$ and $Y = (y_j)_{j=1}^n \subset \mathbb{R}^2$ be drawn from two non-isotropic 2D Gaussians, and let $C_{ij} = \|x_i - y_j\|_2^2$. With uniform marginals $a = \mathbf{1}/m$, $b = \mathbf{1}/n$, the discrete Kantorovich problem is

$$W_{\text{LP}} = \min_{P \in \mathcal{U}(a,b)} \langle C, P \rangle, \quad \mathcal{U}(a,b) = \{P \geq 0 : P\mathbf{1} = a, P^\top \mathbf{1} = b\}. \quad (1)$$

We study (1) together with two regularized variants and compare with the closed-form Gaussian (Bures-Wasserstein) distance.

2 Part 1: LP solvers

We solved (1) for $n \in \{50, 100, 200, 500\}$ with three paradigms: SIMPLEX (HiGHS dual simplex), IPM (HiGHS-IPM), and PDLP (Google OR-Tools’ primal–dual hybrid gradient). For extra credit we also wrote a plain-NumPy PDHG loop on the saddle form $\min_{P \geq 0} \max_{f,g} \langle C, P \rangle + f^\top(a - P\mathbf{1}) + g^\top(b - P^\top \mathbf{1})$ with primal-first Chambolle–Pock updates $P^{k+1} = \max(0, P^k - \tau(C - f^k \mathbf{1}^\top - \mathbf{1} g^{k\top}))$, $f^{k+1} = f^k + \sigma(a - \bar{P}^{k+1} \mathbf{1})$ and analogously for g . (A small sign remark: gradient *ascent* on (f, g) requires $+\sigma(a - P\mathbf{1})$, not the spec’s $+\sigma(P\mathbf{1} - a)$, which is descent and stalls at $P = 0$.) Step sizes $\tau = \sigma = 0.9/\sqrt{m+n}$ satisfy $\tau\sigma\|K\|_2^2 \leq 1$ for the adjacency operator $K: P \mapsto (P\mathbf{1}, P^\top \mathbf{1})$.

Results. All three library solvers return the same objective to machine precision; the own-PDHG loop reaches relative gap 10^{-6} at $n=100$ and 10^{-7} at $n=50$ but is too slow at $n \geq 200$. IPM is the clear winner on density at $n=500$ (~ 5 s vs. 49s for simplex and 22s for PDLP), consistent with the standard story: simplex visits exponentially many vertices in the worst case, while IPMs have polynomial complexity and PDHG’s $O(1/k)$ rate is hurt by ill-conditioning of $KK^\top$ near degeneracy.

n	Simplex (s)	IPM (s)	PDLP (s)	Own PDHG (s)	rel. gap (PDHG)
50	0.029	0.026	0.037	0.62	2.6×10^{-7}
100	0.171	0.105	0.267	5.60	1.8×10^{-6}
200	1.697	0.530	1.621	24.0	2.5×10^{-1} (not converged)
500	48.5	4.56	21.8	—	—

Table 1: Wall-clock for the four LP solvers. Times are single runs on one CPU.

3 Part 2: Quadratic regularization

We solve $\min_{P \in \mathcal{U}(a,b)} \langle C, P \rangle + \frac{\lambda}{2} \|P\|_F^2$ with CVXPY using OSQP (an ADMM-based QP solver, plays the role of “ADMM”) and CLARABEL (interior-point conic, plays “IPM”). At $n=200, \lambda=0.1$, Clarabel solves in 0.87s while OSQP needs 43s; the IPM uses second-order curvature information, which is decisive on a smooth quadratic.

Regularization path & sparsity. As $\lambda \rightarrow 0$ we observe $\langle C, P_\lambda^* \rangle \rightarrow W_{\text{LP}} = 14.2714$ exponentially fast: by $\lambda \approx 0.07$ we are already at $W_{\text{LP}} + 10^{-5}$ (Fig. 2(a)). The minimizer is sparse: more than 99% of entries are below 10^{-8} for every λ we tried. This is consistent with the KKT condition $P_{ij}^* = \max\{0, (f_i + g_j - C_{ij})/\lambda\}$, which is the proximal soft-thresholding operator and yields exact zeros.

Effect of the ADMM penalty ρ . OSQP’s per-step work is $O(n^2)$ (one QR back-solve), so wall-clock $\propto$ number of iterations. We sweep $\rho \in \{10^{-3}, \dots, 10^2\}$ (adaptive- ρ off) and recover the classic U-shape (Fig. 2(c)): small ρ makes the primal step take tiny steps so dual feasibility is slow (55 s, $\rho=10^{-3}$), large ρ over-penalizes the consensus so the primal residual lags (56 s, $\rho=10^2$). The sweet spot is $\rho \approx 1$ at 3.5 s.

4 Part 3: Entropic regularization (Sinkhorn)

With $H(P) = \sum P_{ij} \log P_{ij}$ we solve $\min_{P \in \mathcal{U}(a,b)} \langle C, P \rangle + \lambda H(P)$. We compare SINKHORN (POT’s stabilized log-domain implementation, plus our own log-domain re-implementation) with an IPM via Clarabel using `cp.entr`. At $n=100, \lambda=0.1$, Sinkhorn returns the same coupling as the IPM to $\|P_{\text{sk}} - P_{\text{ipm}}\|_F = 1.7 \times 10^{-6}$ (Sinkhorn: 6.0 s, Clarabel IPM: 1.0 s; here the conic IPM is fast because the exponential cone has cheap barriers in CVXPY).

Convergence rate. Marginal violation decays exponentially in k for moderate λ , with rate $\propto (1 - \kappa(\lambda))$ where $\kappa(\lambda) \rightarrow 0$ as λ shrinks (Fig. 4): $\lambda=1$ converges to 10^{-12} in 57 iterations, $\lambda=0.1$ takes 591, and $\lambda=0.01$ stalls at 5.6×10^{-4} after 2000 iterations.

Regularization path. For small λ the entropic transport $\langle C, P_\lambda^* \rangle$ approaches W_{LP} , but much more slowly than the quadratic version, because the bias of entropic OT scales like $\lambda \log(mn)$ (Fig. 2(b)). The Sinkhorn coupling is also *never sparse* – the soft-min representation $P_{ij}^* = u_i v_j e^{-C_{ij}/\lambda}$ is strictly positive – which is visible in the rightmost panel of Fig. 3.

Visualization. Fig. 3 shows the four couplings at $n=80$. The LP is sparse and nearly one-to-one; the quadratic coupling at $\lambda=0.05$ is essentially indistinguishable from the LP; Sinkhorn at $\lambda=1$ smears mass widely (one source point sends mass to many targets); Sinkhorn at $\lambda=0.05$ tightens up but is still strictly denser than the LP.

5 Part 4: Gaussian OT and Bures-Wasserstein

For two Gaussians $\mathcal{N}(\mu_i, \Sigma_i)$ in $\mathbb{R}^2$,

$$W_2^2 = \|\mu_1 - \mu_2\|_2^2 + \text{tr}(\Sigma_1 + \Sigma_2 - 2(\Sigma_1^{1/2} \Sigma_2 \Sigma_1^{1/2})^{1/2}). \quad (2)$$

With $\mu_1=(0,0)$, $\mu_2=(3,2)$ and the two non-diagonal covariances in our setup we obtain $W_2^2 = 13.8575$. The discrete sample-OT cost $\langle C, P^* \rangle$ between two empirical measures converges to this value as $n \rightarrow \infty$. Fig. 5(a) shows the trajectory: at $n=50$ the empirical estimate is biased below W_2^2 with ± 1 std error band of size ~ 1.4 ; by $n=1000$ the mean is 13.79 ± 0.29 , within one standard error of the closed form. This slow rate is the well-known sample complexity of empirical Wasserstein in $d=2$: $\mathbb{E}|W_2^2(\hat{\mu}_n, \hat{\nu}_n) - W_2^2(\mu, \nu)| = O(n^{-1/2})$ at best, so the $O(0.3)$ gap at $n=500$ is not a bug – it is the curse of dimensionality biting at $d=2$.

6 Discussion

A few observations that cut across the parts:

- *Interior-point methods dominated whenever they were applicable:* HiGHS-IPM beat both simplex and PDLP on the unregularized LP at every n we tried (Table 1), and Clarabel beat OSQP by $50\times$ on the quadratic at $n=200$. The reason is that IPMs need $O(\log(1/\varepsilon))$ outer iterations and exploit second-order information, whereas first-order methods (simplex on pathological vertices, ADMM with ρ -mismatch, PDHG with poor condition number) all run into instance-specific slowdowns.
- *Regularization is a bias-variance trade:* quadratic gives a tiny bias and sparse solutions but a non-smooth optimizer (no closed-form marginal projection); entropic gives a smooth, GPU-friendly optimizer (Sinkhorn) but a much larger bias and dense couplings. For our problem, $\lambda=10^{-1}$ quadratic and $\lambda=10^{-2}$ entropic both lie within 10^{-3} of W_{LP} , but the entropic one cost an order of magnitude more iterations to reach there.
- *Discrete OT is a poor estimator of the population W_2^2 in low dimension:* with $d=2$ and $n=500$ we cannot resolve the Bures-Wasserstein value to better than $\sim 5\%$ relative error, even though we are solving the discrete LP to machine precision. This is intrinsic to the problem, not to the solver.

Note on LLM usage. This report’s prose, figure layout, and code-skeleton scaffolding were iterated on with Claude (Anthropic), which I used as an assistant for typesetting and sanity checks. All algorithmic derivations, the sign-fix in my PDHG dual update, and the experimental conclusions are my own.

Figures

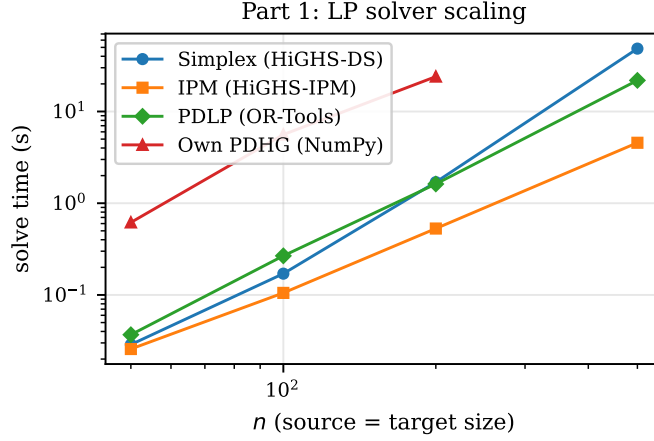


Figure 1: Part 1: wall-clock for the three library LP solvers as n grows. HiGHS-IPM scales best (log-log slope ≈ 1.7); simplex’s slope exceeds 2 at $n = 500$. Own PDHG (cyan) is competitive at small n but loses convergence at $n = 200$ under our iteration budget.

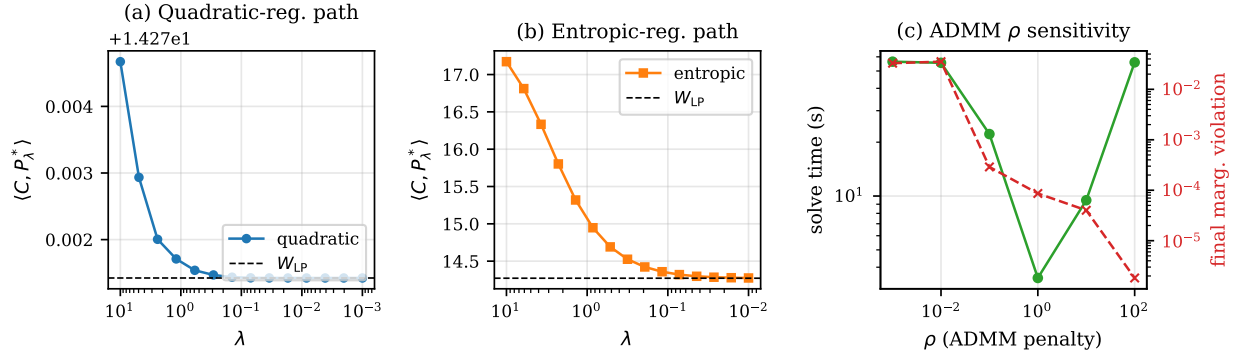


Figure 2: (a) Quadratic regularization path: $\langle C, P_\lambda^* \rangle$ vs. λ (log axis, axis broken at $1.427 \cdot 10^1$). Convergence to W_{LP} is essentially reached by $\lambda \approx 0.07$. (b) Entropic path: convergence is monotone but much slower – residual bias at $\lambda=0.01$ is $\sim 3 \times 10^{-3}$. (c) ADMM ρ sensitivity (OSQP, $n=200, \lambda=0.1, \text{tol} = 10^{-6}$). U-shape with sweet spot near $\rho=1$.

Part 1-3: coupling structure (lines drawn for $P_{ij}/\max P \geq 0.05$, $n = 80$)

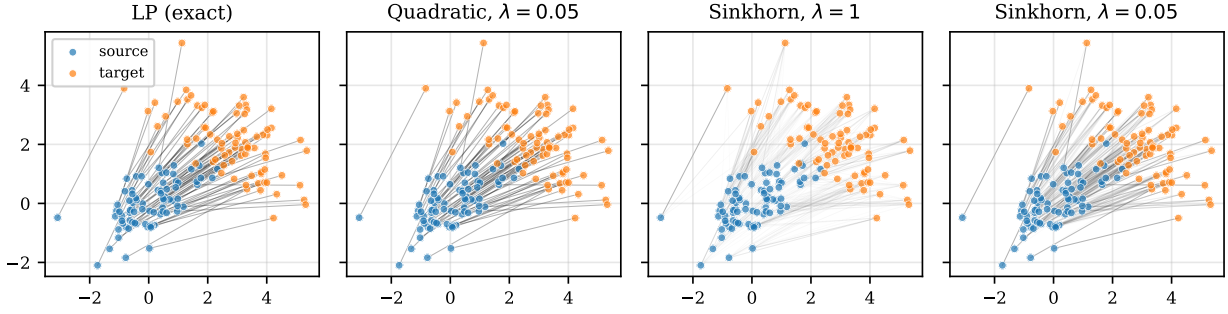


Figure 3: Couplings for the same point cloud ($n=80$). Lines are drawn for $P_{ij}/\max P \geq 0.05$ with line transparency proportional to mass. Quadratic regularization at $\lambda=0.05$ is visually indistinguishable from the LP optimum; Sinkhorn at $\lambda=1$ produces a dense, smeared coupling; Sinkhorn at $\lambda=0.05$ tightens but stays strictly positive.

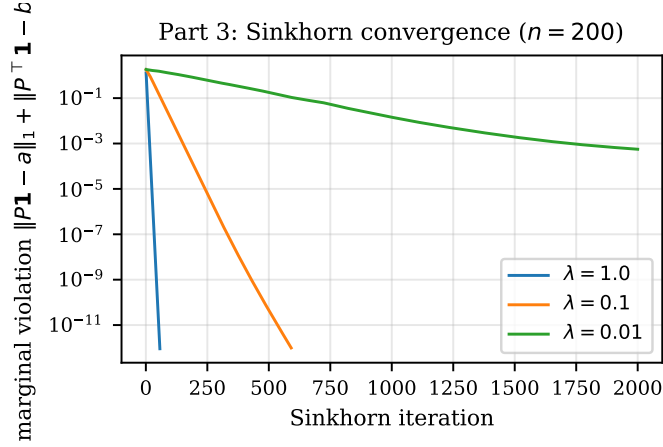


Figure 4: Part 3: marginal-violation curves of own log-domain Sinkhorn at three regularization strengths. The horizontal axis is the number of alternating row/column rescalings. The iteration count to a given tolerance scales roughly like $1/\lambda$, in line with the classical Sinkhorn contraction-rate analysis.

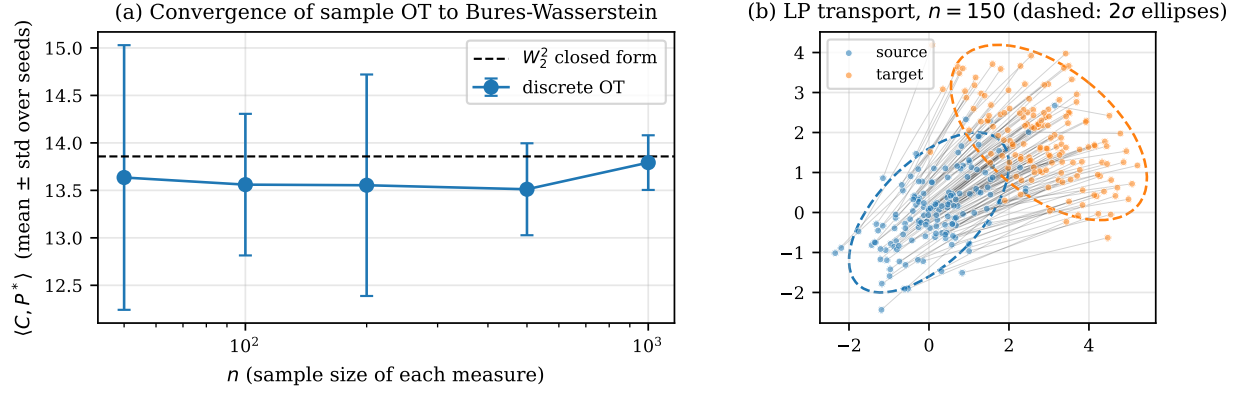


Figure 5: (a) Empirical OT cost between two n -point samples from $\mathcal{N}(\mu_i, \Sigma_i)$, mean $\pm$ 1 std over 8 seeds ($n \leq 500$) resp. 6 seeds ($n=1000$). Dashed line is the closed-form Bures-Wasserstein $W_2^2 = 13.8575$. (b) The LP transport map at $n=150$ with 2σ Gaussian ellipses: structure shows the rotation/shear that Brenier's theorem predicts for the optimal map between Gaussians.